

Programming and Data Structures Laboratory (CS19003)

Assignment 5 (Section 6): 1D array

Department of CSE, IIT Kharagpur

11th March, 2026

General instructions to be followed strictly:

- Name your file as `<roll_no>_<assignment_no>_<question_no>`. For example, if your roll number is 14CS10001 and you are submitting assignment 3 question 1, then name your file as 14CS10001_3_1.c
 - Write your name, roll number, and assignment number at the beginning of your program.
 - Use proper indentation in your code and comment. Marks may be deducted without proper indentation and comments.
 - Submitting the **correct file** is your responsibility, any request of re-submission after the lab will **never** be entertained.
 - **Zero tolerance** policy will be strictly followed for **plagiarism**. The marks will be put straight zero if such cases are identified.
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Question-1 (15 points)

Write two functions `'int reverse(int N)'` and `'int reverse_recursive(int N)'` both of which takes a positive integer N as argument and returns the integer which is formed by reversing the order of the digits of N . In `'int reverse(int N)'`, do not use recursion to get the reversed number while in `'int reverse_recursive(int N)'` recursively call the function to get the reversed number. If required, you can use a global variable [that can be used throughout the program]. Write a `main()` function to demonstrate use of both the functions.

Example:

```
Please provide the positive integer to be reversed: 1234
The reverse of 1234 using 'reverse_recursive' is: 4321
The reverse of 1234 using 'reverse' is: 4321
```

```
Please provide the positive integer to be reversed: 7890
The reverse of 7890 using 'reverse_recursive' is: 987
The reverse of 7890 using 'reverse' is: 987
```

Question-2 (15 points)

Write a function `'int MaxMinDiff ( int arr[ ], int n )'` that takes an array of n integers as argument, and returns the difference between the maximum and minimum integers in the array. Write a `main( )` function that takes a list of integers as input from the user, stores the integers in an array and then uses the function to print the difference between the maximum and minimum integers in the list. You can take the maximum size of the array as 100. Example interaction with the program:

Example:

Enter the number of integers: 6
Enter the integers separated by space: 42 20 31 76 100 82
Difference between max and min: 80

Question-3 (20 points)

You are given an array of N non-negative integers. You start at index 0. Your goal is to determine if you can reach the last index ($N - 1$) of the array by following these movement rules.

- At any index i , the value stored in the array at that position, (let's call it $K = arr[i]$), represents the jump distance.
- From index i , you can either jump forward or jump backward, *i.e.*,
 - (This is tried first): $i + K$
 - (If the above leads to dead end (explained below), this is tried): $i - K$
- You cannot move to an index that is outside the bounds of the array (less than 0 or greater than or equal to N).
- To prevent getting stuck in an infinite loop (jumping back and forth between the same two indices), you must keep track of where you have already been.

Write a **recursive** function '`int canReach(int arr[], int visited[], int n, int current)`' that returns 1 if the last index is reachable from the current position and 0 otherwise. Write a `main( )` function that takes a list of integers as input from the user, stores the integers in an array and then uses the function to print if the last index is reachable from the first index using the movement rules. The array '`visited`' initialized to all zeros is needed to keep track of the visited indices. Before jumping to a new index, mark the current index as visited. You can take the maximum size of these arrays to be 100.

The figure below shows an illustration with an array containing the values: 2, 4, 1, 0, 5, 2.

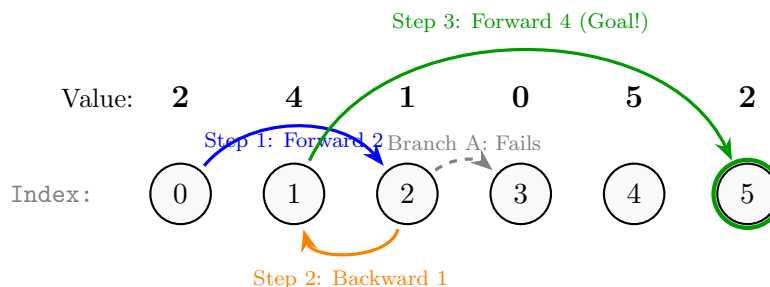


Figure 1: Nodes represent the array indices (0 to 5). Numbers above nodes represent the “jump distance” at that index. Solid arrows mean successful path and dashed arrow means dead-end.

Hint: You have to keep the following in mind.

- Base cases: Jumping out of bounds, Having reached the goal, checking already visited index (this prevents infinite loop)
- You need to set the current index as visited in the array `visited`
- First you should try jumping forward, then backward

Example:

Enter number of elements in the array: 6
Enter array elements separated by space: 2 4 1 0 5 2
Result: The last index is reachable.

Enter number of elements in the array: 5
Enter array elements separated by space: 1 1 0 1 1
Result: The last index is not reachable.